

Small Language Model Research

CSE299 Junior Design — Section 04, Group 03

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I. INTRODUCTION

This week's research explored three interconnected questions in efficient and trustworthy AI: whether SLM guidance can reduce false confidence in reasoning pipelines, whether model compression can enable privacy-safe local deployment in healthcare, and whether small-scale fine-tuning genuinely improves or degrades pre-trained models.

II. WORK COMPLETED

A. SLM Guidance vs. False Confidence: A GSM8K Evaluation:

- Implemented a dual-mode evaluation pipeline comparing a fine-tuned SLM-guided system against an unguided baseline on GSM8K, producing structured results across 100 questions per condition including per-vote correctness tracking and confidence scores.
- Identified a significant false confidence problem in the unguided baseline: 68 questions triggered high vote-agreement (≥ 0.8) but only 10.3% were correct, compared to 51.0% accuracy under the same confidence threshold in the guided pipeline — suggesting guidance suppresses convergence on wrong answers.
- Observed a 1.82× improvement in vote consistency under guidance, with 26 questions reaching high reliable agreement (80–100% of votes correct) versus only 7 in the baseline — indicating the fine-tuned guide improves per-attempt reasoning quality, not just final answer selection.
- Discovered an accuracy inconsistency between two separate runs (56% vs 34% guided accuracy) which is suspected to be caused by random seed variation in question sampling.
- Defined three measurable research angles — compute efficiency, vote consistency, and confidence calibration.

B. Compressing SLMs for Local Healthcare: Benchmarks & Baselines:

- Added Penn Treebank (PTB) dataset alongside WikiText-103 to prove generalization.
- Recognized missing baselines: GPTQ, LoRA, SparseGPT

must be added.

- Problem: LLM-based healthcare tools send patient data to the cloud which is a legal violation.
- Solution: Started building a privacy-first local healthcare chatbot using compressed SLMs.

C. Does Fine-Tuning Actually Help?

- Fine-tuned two LLMs (Gemma 3 1B and Phi-4 Mini) on MMLU (general knowledge) and HumanEval (coding) benchmarks using QLoRA — a memory-efficient technique that runs on a free Kaggle P100 GPU.
- Tested data efficiency by training with small sample sizes (50, 200, 500, 1000) to investigate how much fine-tuning data is actually needed to improve model accuracy.
- Found that fine-tuning with limited data consistently hurt accuracy across all 3 experiments — a phenomenon known as catastrophic forgetting, where narrow training overwrites broad pre-trained knowledge.
- Confirmed that model size doesn't prevent degradation — Phi-4 Mini started much stronger (58.2% vs 30.4%) but showed the same declining pattern after fine-tuning, just like the smaller Gemma model.
- Concluded that pre-training quality matters more than small-scale fine-tuning, and that meaningful improvement requires 10K+ well-curated samples — while validating QLoRA as a practical, zero-cost approach accessible to anyone.

III. CONCLUSION

Across all three tracks, a clear theme emerged — scale and quality of both data and guidance matter more than shortcuts. SLM guidance meaningfully suppresses false confidence, compression offers a viable path to local healthcare AI, and small-scale fine-tuning consistently backfires, reinforcing that robust AI requires careful, well-resourced training rather than quick interventions.

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